# Fat-Sat

- Fat-Sat Methods
- Cause and effect
- Where to begin
- What to control and watch out for
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- Q&A

# Fat Saturation Methods

Two methods:

- **Chemical Shift Imaging (CSI)**
  - **Frequency Selective Excitation**
  - **Phase Sensitive**

- **T1 Relaxation Method**

# *CSI Methods*

Frequency Selective Excitation

- **Easy to implement – wide range of applications**
- **Susceptible to local $B_0$ & $B_1$ fields**
- **Been using it for years**

Phase Sensitive – phase contrast methods (3-point Dixon)

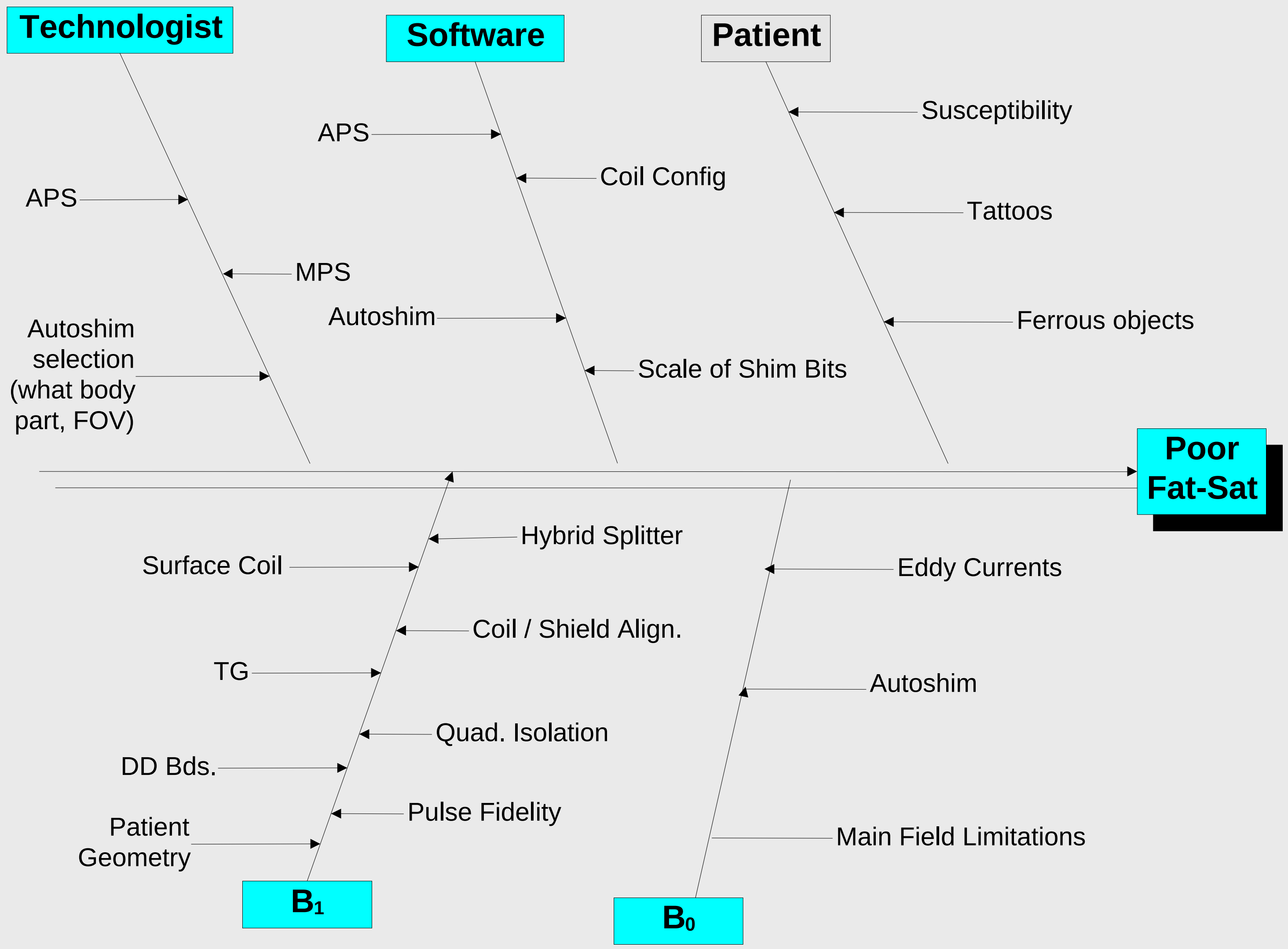
- **Less susceptible to field in-homogeneities**
- **Longer scan times, processing intensive**

# T1 Relaxation Methods

STIR

- **Mid-field & low-field scanner**
- **Fairly insensitive to B0 inhomogeneity**
- **Limits contrast optimization (IR pulse set to null fat)**

# Cause & Effect



# Where to begin….

- Review all slices in the series – note size of patient, weight, and scan parameters used.
- Check the range (too far from iso-center?)
- If shading or abnormal uniformity is seen in fat sat images, reformat the series and examine the anatomical characteristics above and below the shaded slices.
- Look for susceptibility issues due to curves or folds in the anatomy that are adjacent to the shaded areas.
- Check the center frequency of the fatsat series against other series in the same plane.
- Check the TG value for the fatsat series against other series in the same plane.
- Check IQ for other series on the same patient.

# Things you control, that effect Fat-Sat

- **Shim (Supercon and Gradshim)**

✓ **If it's wrong, make it right**
✓ **If it's good, look at B1 homogeneity**
✓ **Look at the coils (e.g. if there's no fat-sat problem in the head, it might be a coil issue)**

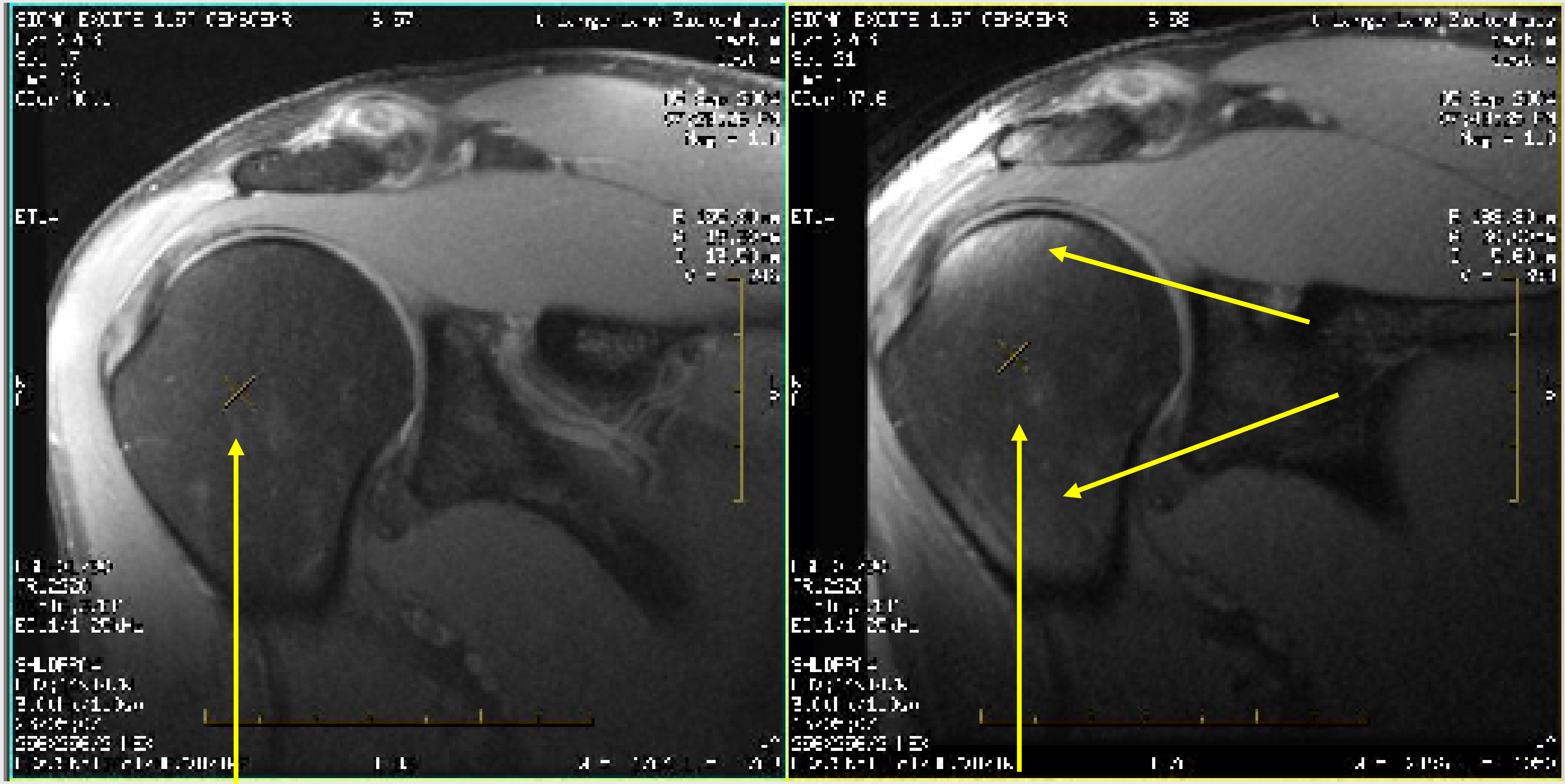
- **Eddy current compensation.**

# *Physics & physical (uncontrolled) phenomena that effect Fat-Sat*

- **Foreign ferrous objects (easily detected and corrected).**
- **Metallic implants (easily detected).**
- **Patient size (if anatomy extends beyond normal 45cm spherical shimming volume).**
- **Local shim disturbances due to susceptibility (difficult to detect and minimize).**

# *Too far from iso-center (1)*

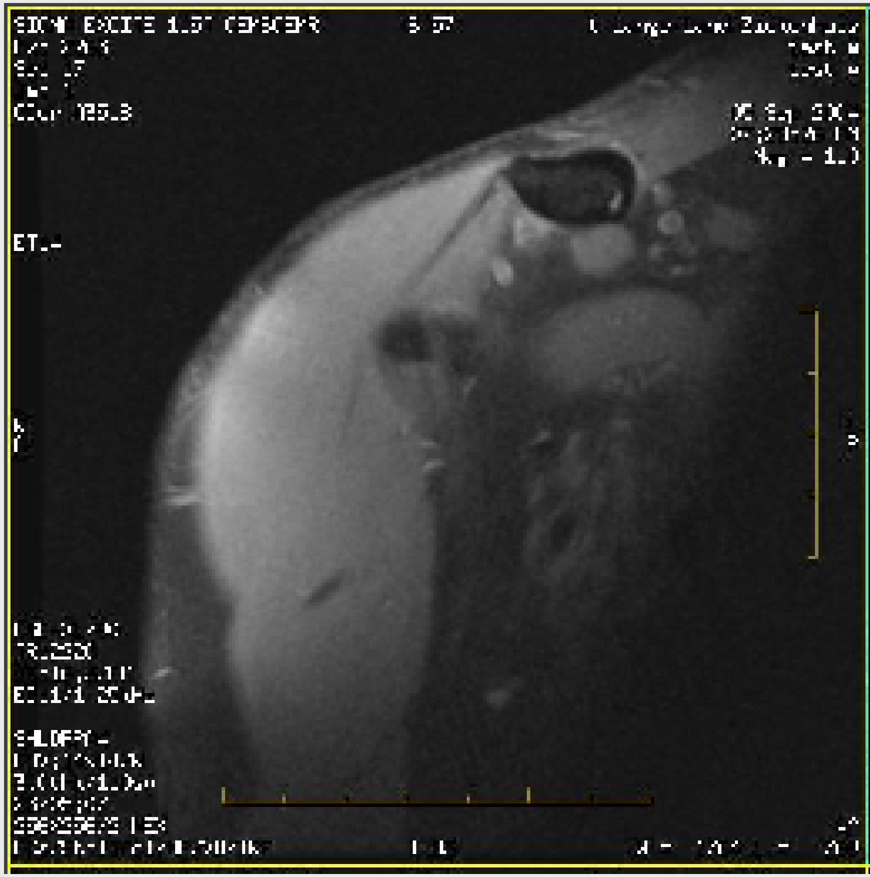
Same large volunteer and protocol, scanned at 2 different locations.



R156 R199

# Too far from iso-center (2)

Stay away from bite-marks, they can effect fat-sat.



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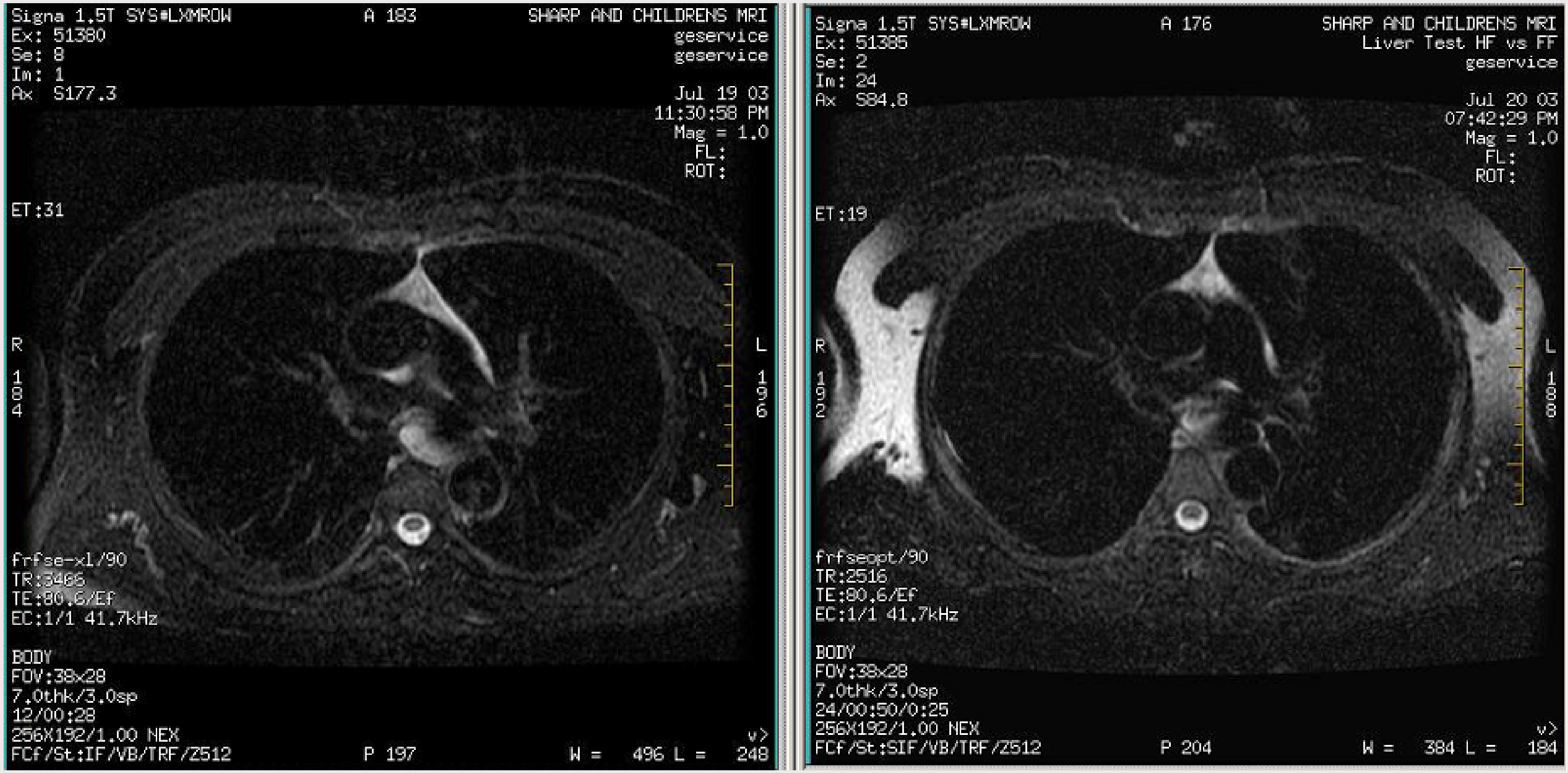
# *Too far from iso-center (3)*

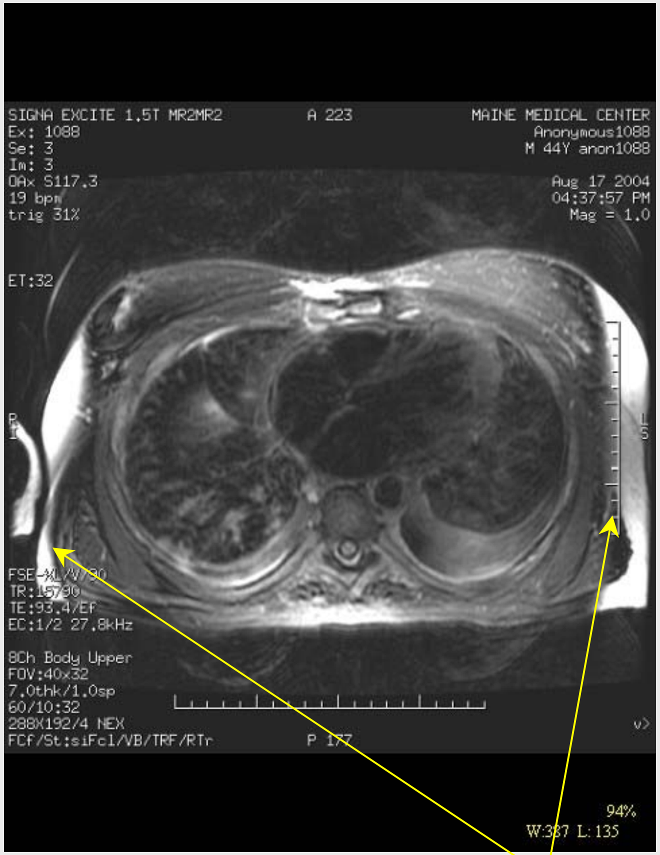
Same volunteer scanned at 2 different locations.



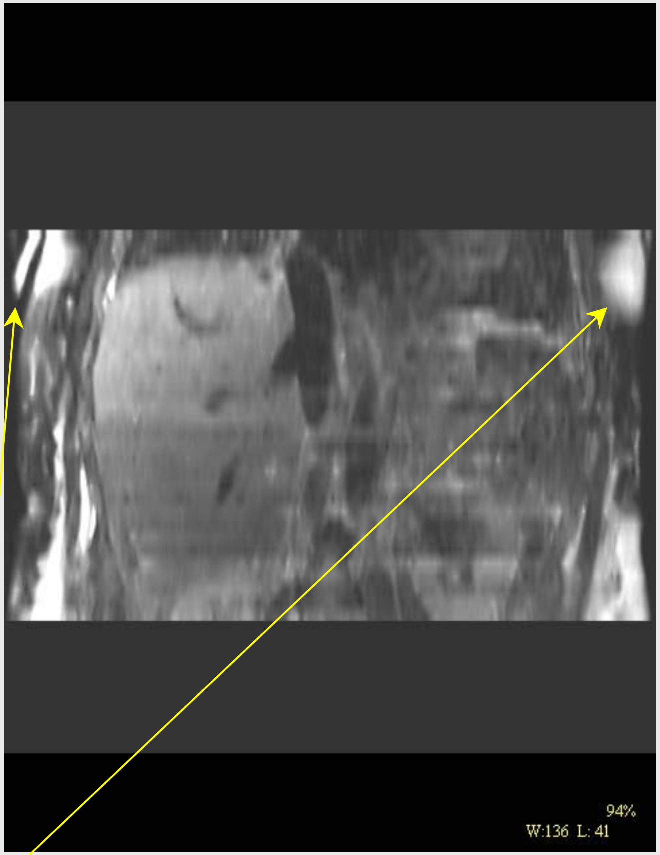

After an upgrade, could acquire 24 images with one scan, one breath-hold, which double the scan range in the S/I direction.

# Too far from iso-center (4)

Fat-Sat in Chest Region




Coronal Reformat of Axial Series

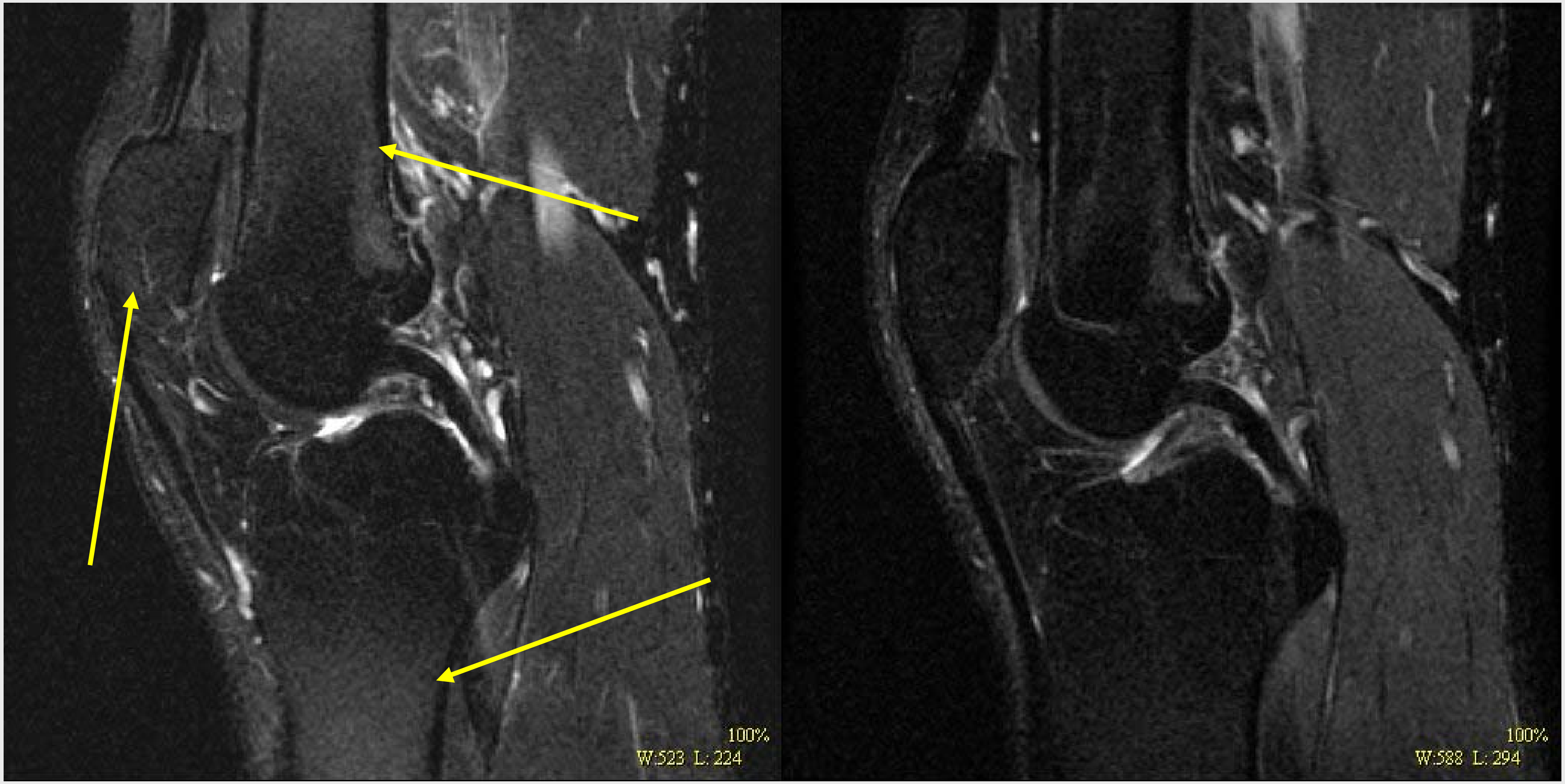



Coronal image from same patient shows anatomy extending into areas where magnetic field is non-uniform.

(Shim is typically very uniform within a 40cm DSV from isocenter)

# Coil problems and fat-sat (1)

Same volunteer and protocol, scanned with 2 different Quadknee coils.




Original Coil

Replacement Coil

# Coil problems and fat-sat (2)

For coil specific problem, substitute other coil and observe problem area.

These tests are what lead to the suspicion that the Quadknee (pervious slide) was causing problem with fat-sat.


Head Coil

GPFLEX Coil

# Coil problems and fat-sat (3)

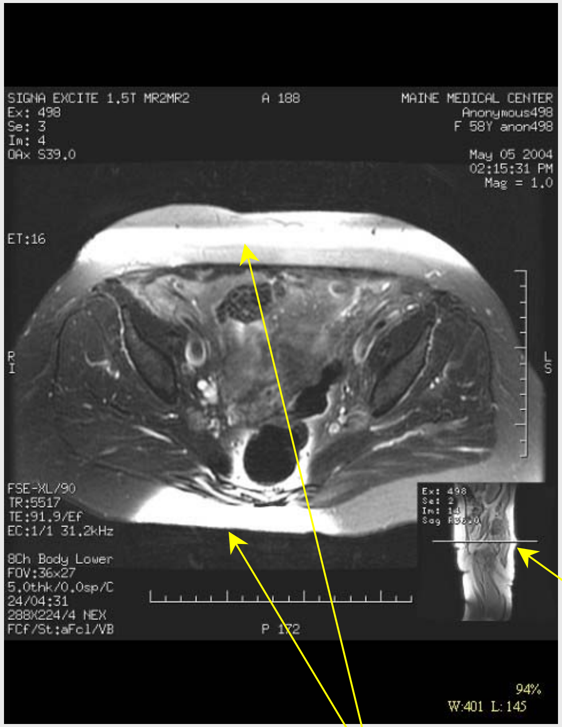
Diagnosed bad element in HRWRIST coil with saline bag and saveinter.

Original Coil

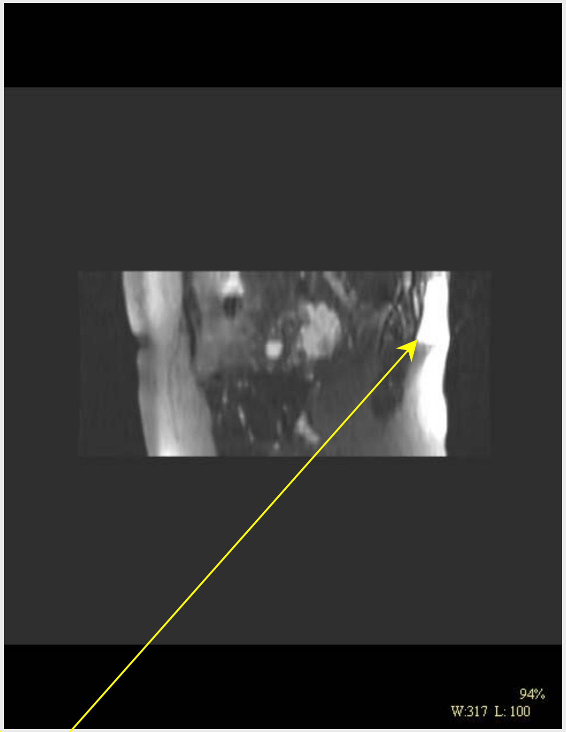
Replacement Coil

# *Susceptibility effects - when magnetic flux lines transition from air to tissue (1)*

Fat-Sat non-uniformity
anterior and posterior




Reformat of Axial Series

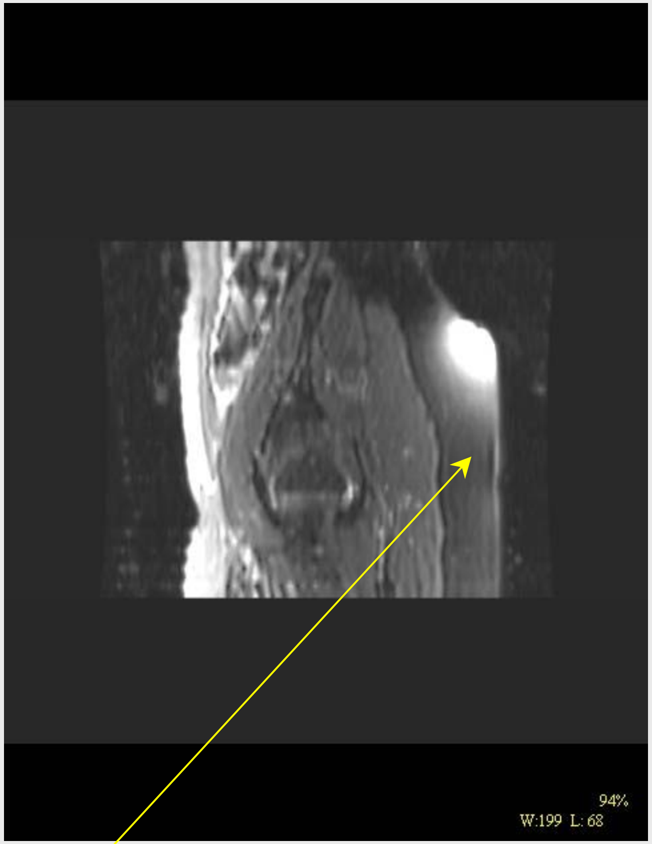



Reformatted image shows local distortion of magnetic field on posterior aspect of patient due to metallic object, and multiple folds in skin on anterior aspect of patient.

# *Susceptibility effects - when magnetic flux lines transition from air to tissue (2)*

Fat-Sat Significant Shading


Reformat of Axial Series

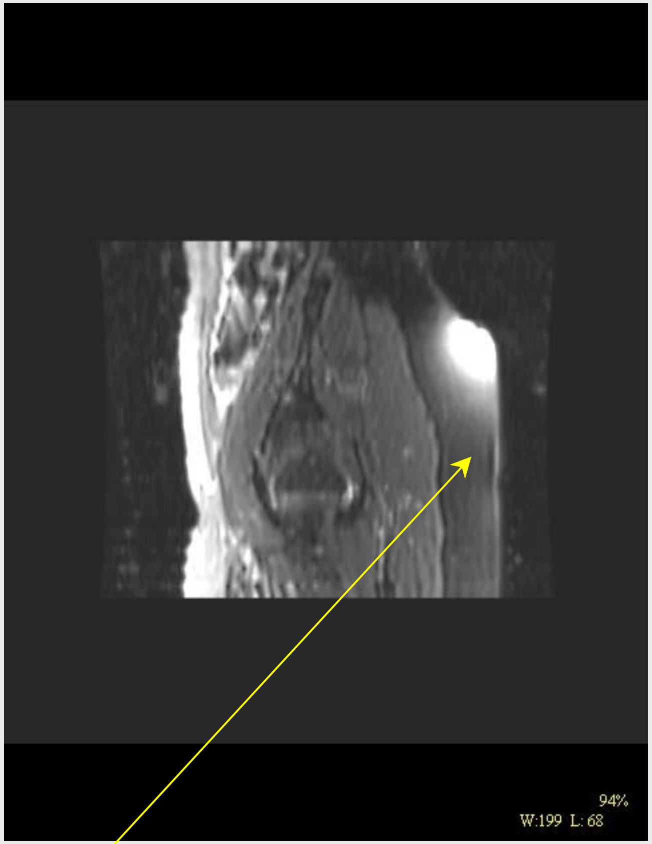



Reformatted image shows area of curvature in anatomy causing a local disturbance in the magnetic field

# *Susceptibility effects - when magnetic flux lines transition from air to tissue (3)*

T2 FatSat


Reformat of Axial Series

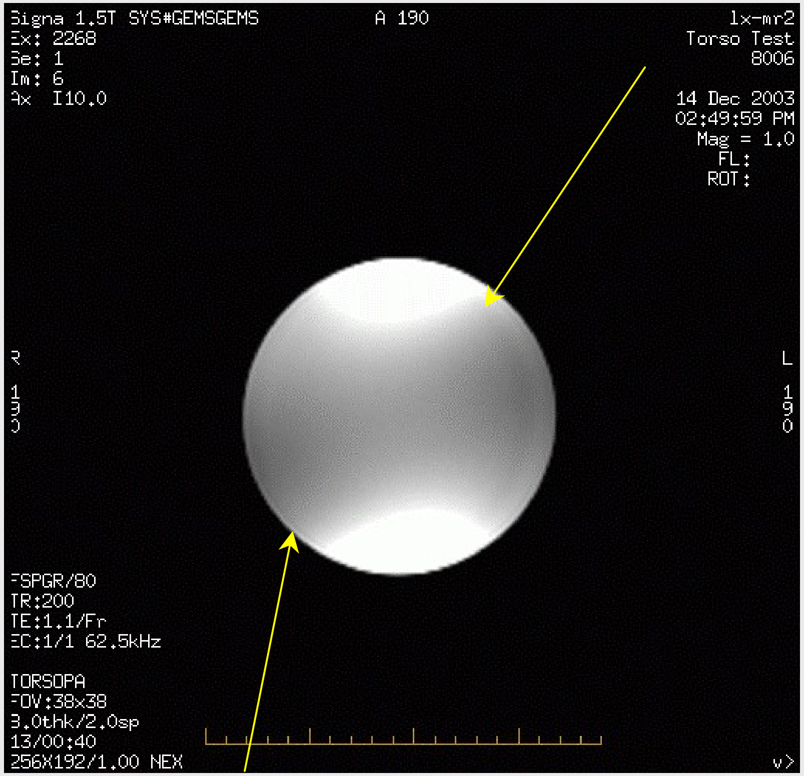



Sagittal image from same patient shows same disturbances near deep curvature in waist area.

# *Normal amounts of shading…(1)*

Asymmetrical shading seen in most patient and phantom scan.

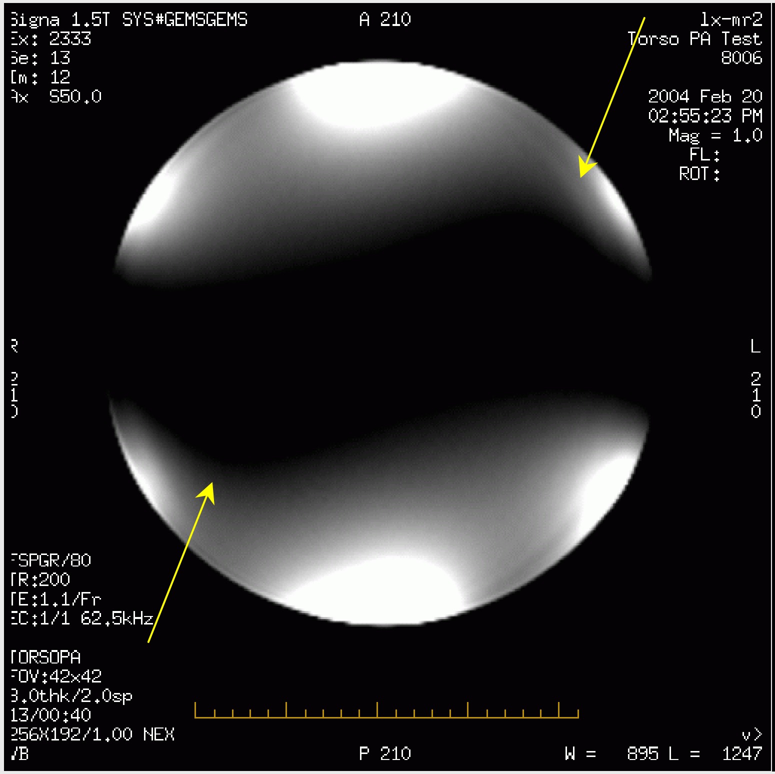

Body Array



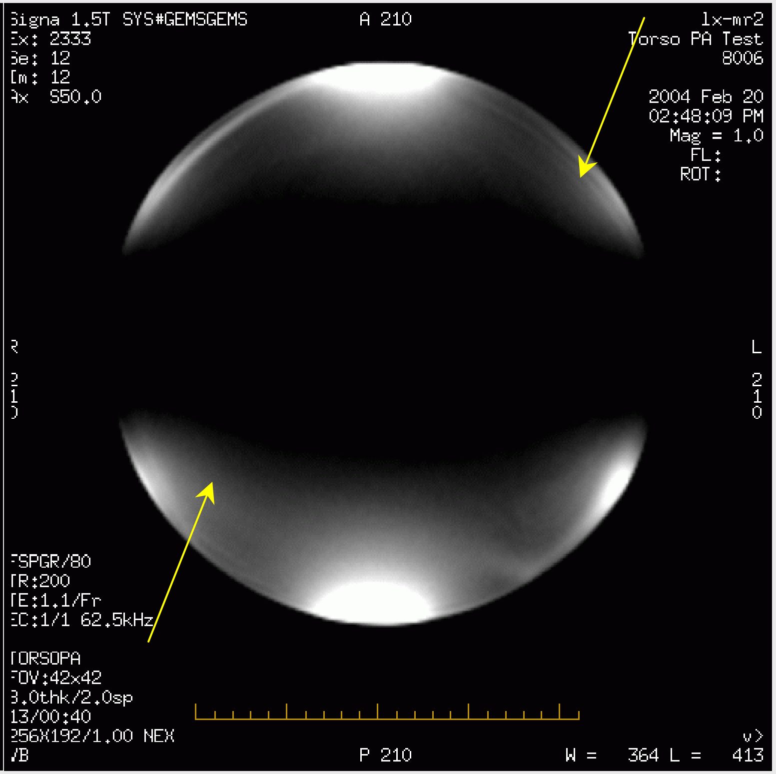

Torso Array

# *Normal amounts of shading…(2)*

Standing wave and dielectric effects (even at 1.5T) and contribute to shading and incomplete fat-sat




CuSO4
Axial Image
TORSOPA

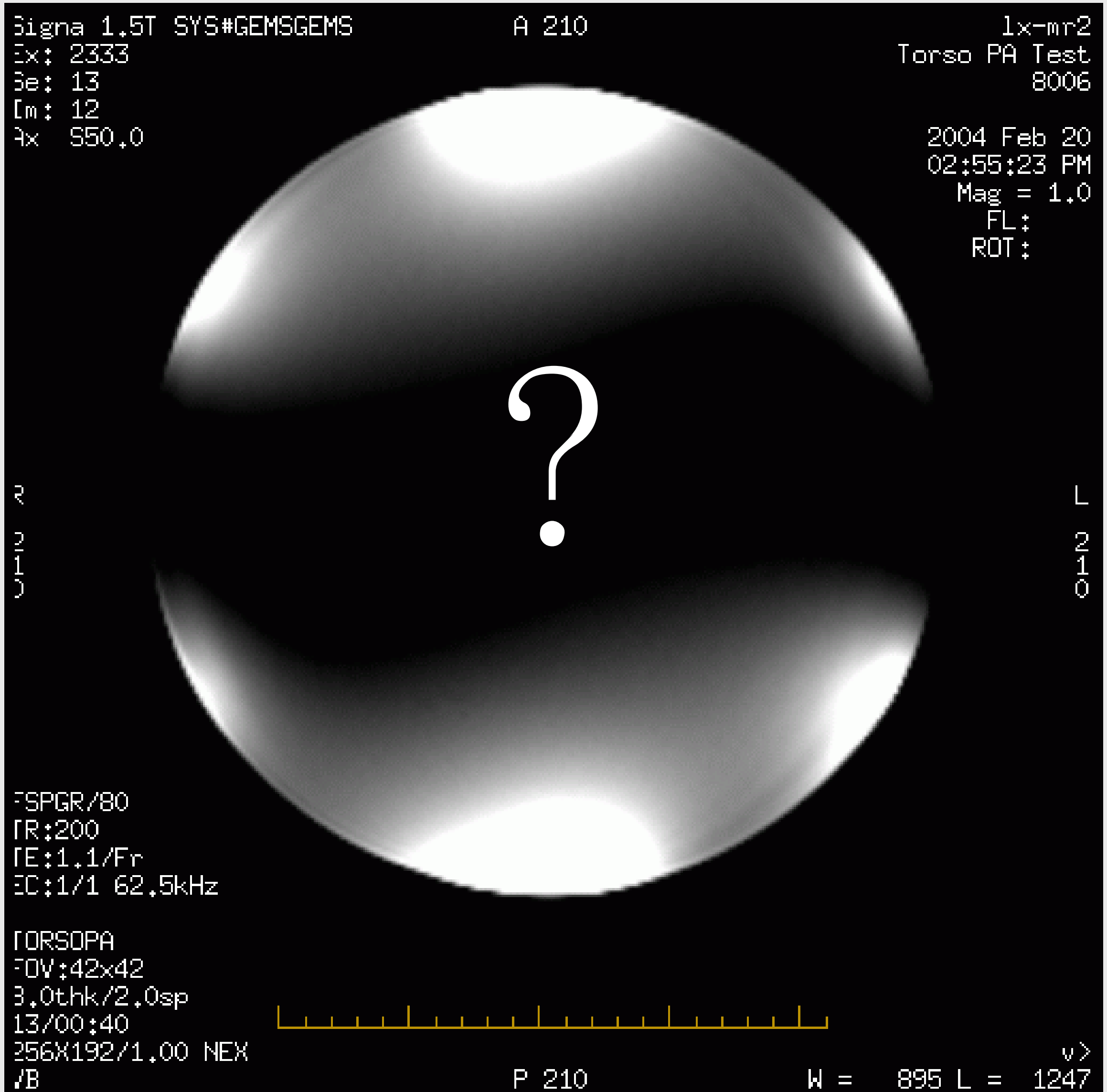



Silicon Oil
Axial Image
TORSOPA

# Summary

- Start any investigation by making sure shim and eddy currents are good.
- Make sure scans are done within a reasonable distance from iso-center.
- Take into consideration the patients impact to the $B_0$ & $B_1$ fields.

# Questions



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Torso PA Test
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m: 12
x S50.0
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